

A CNN-Based Method for Intent Recognition Using Inertial Measurement Units and Intelligent Lower Limb Prosthesis

Ben-Yue Su^{ID}, Jie Wang, Shuang-Qing Liu, Min Sheng, Jing Jiang, and Kui Xiang

Abstract—Powered intelligent lower limb prosthesis can actuate the knee and ankle joints, allowing transfemoral amputees to perform seamless transitions between locomotion states with the help of an intent recognition system. However, prior intent recognition studies often installed multiple sensors on the prosthesis, and they employed machine learning techniques to analyze time-series data with empirical features. We alternatively propose a novel method for training an intent recognition system that provides natural transitions between level walk, stair ascent/descent, and ramp ascent/descent. Since the transition between two neighboring states is driven by motion intent, we aim to explore the mapping between the motion state of a healthy leg and an amputee's motion intent before the upcoming transition of the prosthesis. We use inertial measurement units (IMUs) and put them on the healthy leg of lower limb amputees for monitoring its locomotion state. We analyze IMU data within the early swing phase of the healthy leg, and feed data into a convolutional neural network (CNN) to learn the feature mapping without expert participation. The proposed method can predict the motion intent of both unilateral amputees and the able-bodied, and help to adaptively calibrate the control strategy for actuating powered intelligent prosthesis in advance. The experimental results show that the recognition accuracy can reach a high level (94.15% for the able-bodied, 89.23% for amputees) on 13 classes of motion intent, containing five steady states on different terrains as well as eight transitional states among the steady states.

Index Terms—Intent recognition, lower limb prosthesis, inertial measurement unit (IMU), convolutional neural networks (CNN), swing phase.

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I. INTRODUCTION

SINCE lower limb amputation is a major cause of disability, millions of transfemoral amputees over the world are suffering from difficulty in moving. The presence of prosthetic legs can help transfemoral amputees restore their locomotion and enable them to perform diverse daily activities [1]. However, using passive prosthesis may significantly impair the walking symmetry and metabolic energy efficiency of transfemoral amputees [2], [3]. The problem thereby stimulates a line of research on powered lower limb prosthesis with the aid of robotic techniques [4].

The powered intelligent prosthesis can automatically calibrate torque according to the gait speed and joint angle [5], [6]. A hierarchical control system allows for better control of the prosthesis [7]. As shown in Fig. 1, the hierarchical general control system generally consists of three components. The high-level controller comprehends human motion intent through activity mode recognition or direct volitional control while the middle-level controller perceives intent from the high-level controller and adjusts the control strategy. Then, the low-level controller eventually actuates the prosthesis based on the control strategy [4]. One major issue of research on intelligent lower limb prostheses is motion intent recognition, which influences the effectiveness of the high-level controller. The purpose of motion intent recognition is to accurately interpret the human intent information encoded by signals in the nerve center. An intent recognition algorithm with high performance makes the transitions between two neighboring locomotion modes become seamless, automatic, and natural [8].

The most commonly used sensors for intent recognition cover surface electromyogram (sEMG) sensors and mechanical sensors. Motion intent recognition based on the sEMG was first reported in [9]. Many studies then use pattern recognition and machine learning techniques to analyze the locomotion states and transitions of lower limb amputees. Huang *et al.* [10] first combined sEMG signals with pattern recognition to classify seven daily motion states. They collected signals from eight able-bodied subjects and two subjects with transfemoral amputations while they were walking on different terrains. Moreover, two gait events were proposed (toe-off and heel-strike) to obtain the size of the analysis window. Linear discriminant analysis (LDA) and an artificial neural

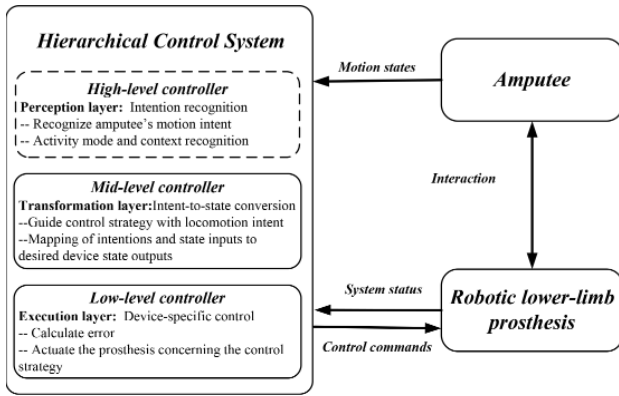


Fig. 1. Hierarchical control strategy of intelligent lower limb prostheses.

network (ANN) were used as classifiers, and the recognition accuracy reached 85.7% [10]. They also used a combination of sEMG and load cell information from five transfemoral amputees [11]. The method could recognize six locomotion modes and five transitions with a high performance ($> 95\%$) on support vector machines (SVMs). Furthermore, they investigated the effects of locomotion mode recognition errors on the volitional control of lower limb prosthesis, and proved that not all of the mode recognition error affected gait stability in prosthesis users. They also defined and suggested “critical errors” as a new index to evaluate locomotion mode recognition [12]. Liu *et al.* [13] used three different adaptive pattern classifiers for locomotion mode recognition, and they indicated that the adaptive classifier could effectively maintain or restore the performance of locomotion mode recognition. Nevertheless, sEMG-based systems for prosthesis usually have poor performance in clinical trials, due to the presence of electrode shift, including the effects of nerve atrophy, variations in muscle contraction effort, and changes in electrode position [14].

Since mechanical signals directly record kinematics information, the studies that used mechanical sensors stand out from other research on intent recognition. In order to realize real-time recognition, Liu *et al.* [14] proposed to solely use a mechanical sensor system consisting of an accelerometer, gyroscope, and two pressure sensors. The intra-class correlation coefficient (ICC) was first calculated according to the Cartesian product of walking speed and terrain. Then, the sensor data were fused via Dempster–Shafer theory and classified with hidden Markov models (HMMs). They successfully predicted prosthesis users’ intent with a rate of 95.8% while the users were walking on different terrains (level ground, ascending stairs, descending stairs, and up/down ramps) [14]. However, the method does not take the transitional states into consideration. Young *et al.* [15] built an intent recognition system using mechanical sensors only (six-axis inertial measurement units (IMUs) and axial load). They collected steady-state and transitional data from six transfemoral amputees while five locomotion modes were performed. The system eventually recognized five steady states as well as eight transitional states, with an overall rate of 93.9%. In their following works, dynamic Bayesian networks (DBNs) combined with

time history information were used to improve the recognition accuracy to 94.7% [16]; moreover, a user-independent intent recognition system was developed to reduce the training burden [17]. Furthermore, an ANN was used with a scaled conjugate gradient learning algorithm to recognize the motion intent of amputees, and suitable and accurate results on off-line classification were acquired [18].

To summarize, prior studies usually used the mechanical or sEMG sensors that were embedded on the prosthesis for data collection, and selected handcrafted features for classification. However, manually selecting available features needs expert participation and prior knowledge. A subjective experience will lead to unavoidable biases and limit the range of system application. Besides, conventional intent recognition methods often involve sensor fusion, which might increase complexities when considering multi-modal sensor signals of transitional states. The goal of this paper is to overcome these drawbacks. Therefore, we adopt a new implementation for sensor data collection, along with a novel redefinition of the transitional state.

In this paper, we propose a new method for training an intent recognition system that provides natural transitions between level walk, stair ascent/descent, and ramp ascent/descent. To reduce the computation cost, we solely use IMUs. Unlike the implementation of a conventional intent recognition system, we place three IMUs on the healthy leg of lower limb amputees, and analyze time series data from IMUs within the early swing phase. The setup allows us to monitor the locomotion of the healthy leg while exploring the mapping between its motion state and the transfemoral amputee’s motion intent. Through the mapping, we expect that the upcoming motion state of the prosthesis can be predicted before the transition driven by motion intent occurs. To avoid manual feature selection, the convolutional neural network (CNN), which has made great strides in computer vision [19]–[21] and signal processing [22], is used in this paper. We employ the CNN to automatically learn the hybrid features without expert participation. Experimental results shows that the method can predict the motion intent of unilateral amputees, including the upcoming locomotion mode as well as the transition between two neighboring states.

The structure of this paper is as follows. Section 2 describes the related works on the CNN and our method in detail. Section 3 illustrates the experimental setup and results. Finally, concluding remarks and discussion are given in Section 4.

II. PROPOSED METHOD

This section first presents the concepts of the steady state/step and transitional state/step, which were first proposed in our previous work [23]. Their definitions can help readers comprehend the purpose and implementation of this paper. Second, we introduce the basic concepts of CNNs, and we review related works on motion state recognition based on CNNs. Additionally, we present the modified architecture designed for intent recognition. The mathematical model is also given to interpret the mechanism for processing intent signals.

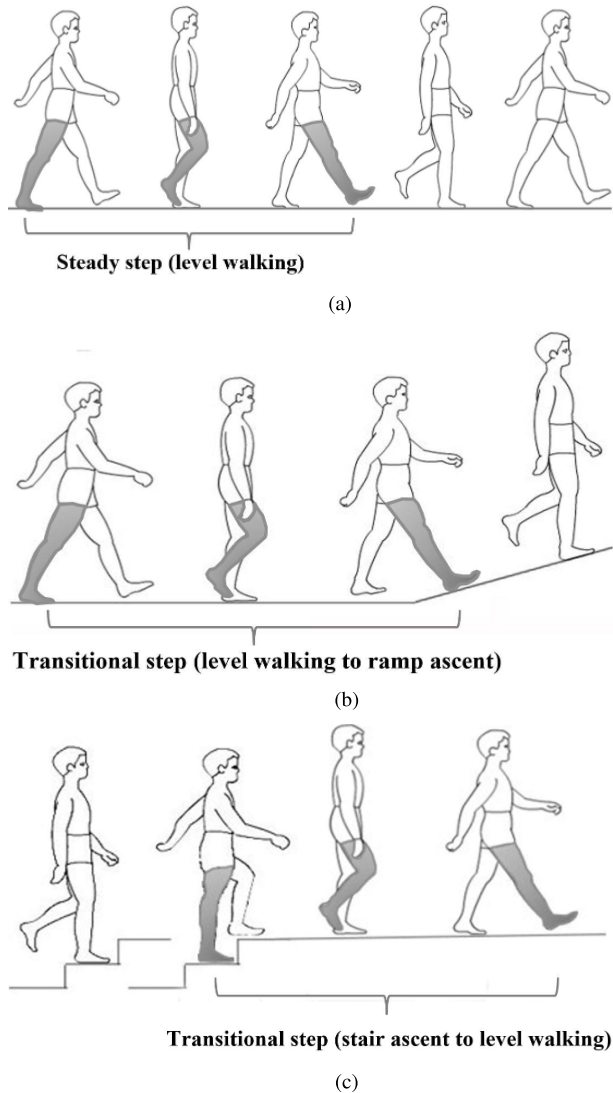


Fig. 2. Illustration of steady step transitional step. (a) Steady step. (b) Transitional step (from level ground to a ramp/stairs). (c) Transitional step (from a ramp/stairs to level ground).

A. Definitions of Motion States

The motion states conducted by the lower limb has an inherent regularity. The regularity is often investigated within a single gait cycle, which consists of a swing phase and a stance phase [24]. Based on the principle, we put forward some new definitions of the motion states within a single gait cycle [23]. Furthermore, the steady and transitional states are exactly expressed by steady and transitional steps performed in the swing phase. The relevant definitions are given below. Figure 2 also illustrates the steady and transitional steps for better interpreting the proposed method.

The steady step begins at toe-off and ends with a heel-strike. The procedure within the swing phase is depicted in Fig. 2(a).

Second, as seen from Fig. 2(b), the transitional step begins at toe-off, where the foot rises from the level ground and continues until the heel touches the ramp or stair.

To express the transition step that starts from a ramp ascent/descent and ends at level ground, a third definition

is provided. As shown in Fig. 2(c), the transitional step starts at toe-off, while the contralateral side begins to stand at the level ground, and ends at the heel-strike.

There are 13 motion states investigated in this study (five steady states and eight transitional states among the steady states on five different terrains): *level walking (LW)*, *stair ascent (SA)*, *stair descent (SD)*, *ramp ascent (RA)*, *ramp down (RD)*, *level walking to stair ascent (LW-SA)*, *level walking to stair descent (LW-SD)*, *level walking to ramp ascent (LW-RA)*, *level walking to ramp descent (LW-RD)*, *stair ascent to level walking (SA-LW)*, *stair descent to level walking (SD-LW)*, *ramp ascent to level walking (RA-LW)*, and *ramp descent to level walking (RD-LW)*.

Note that we focus on the transitional step performed by the healthy leg for exploring the mapping between the transitional state/step and amputees' intent. Thus the subjects have to adjust their step length to make sure the transitions occur at the swing phase of the sound leg (the details of the step sequence are shown in Table I). Through perceiving the current motion state of the healthy leg, the embedded intent recognition system can first analyze the amputee's intent and then modify the control strategy in advance for providing natural and seamless transitions. Besides, transitions are constrained to occur at the stance phase of the affected side in real application. This can be viewed as a safety measure to eliminate potential hazards associated with changing motion states while the prosthesis is swinging or approaching the ground.

B. CNNs for Intent Recognition

CNN has received significant attention, leading technological advances in various domains: image classification [25], semantic segmentation [26], and object detection [27]. The CNN architecture was historically inspired by the structure of the mammalian visual system [28], *i.e.*, the hierarchical structure of the visual cortex [29]. The engineering success of the CNN compared to the fully connected neural network is largely due to the strong regularization imposed by the convolution structure and pooling layer: with far fewer weights to learn, the network can generalize better [30].

Due to the attractive qualities of the CNN, and the relative efficiency of its local architecture, the network has been proven to be particularly powerful when addressing signals and time series, given the ability of this type of network to automatically extract local features from raw data while remaining aware of the data structure [31]. It is commonplace to employ CNN for motion state recognition, *i.e.*, human activity recognition (HAR) [32], especially HAR based on wearable sensors [33]–[35]. Generally, HAR based on wearable sensors aims to classify multiple classes of human motion states through multichannel sensor signals. Moreover, some previous studies even realized automatic recognition, which in essence benefits from the hierarchical structure of the CNN. This line of study stimulates our research on intent recognition.

Traditional intent recognition studies predefined a set of handcrafted features based on the prior knowledge, and selected a subset of features by a trial and error method. To this end, we therefore design a CNN structure to automatically learn the high-level and mid-level features from the sensor

TABLE I
MOTION STATES AND CORRESPONDING STEP SEQUENCE

Motions	Details of step sequence
Five steady states	Sound leg – affected side – sound leg – affected side – sound leg
Level walking to stair ascent (LW-SA)	Sound leg (LW) – affected side (LW) – sound leg (LW-SA) – affected side (LW-SA) – sound leg (SA)
Level walking to stair descent (LW-SD)	Sound leg (LW) – affected side (LW) – sound leg (LW-SD) – affected side (LW-SD) – sound leg (SD)
Level walking to ramp ascent (LW-RA)	Sound leg (LW) – affected side (LW) – sound leg (LW-RA) – affected side (LW-RA) – sound leg (RA)
Level walking to ramp descent (LW-RD)	Sound leg (LW) – affected side (LW) – sound leg (LW-RD) – affected side (LW-RD) – sound leg (RD)
Stair ascent to level walking (SA-LW)	Affected side (SA) – sound leg (SA) – affected side (SA-LW) – sound side (SA-LW) – affected side (LW)
Stair descent to level walking (SD-LW)	Affected side (SD) – sound leg (SD) – affected side (SD-LW) – sound side (SD-LW) – affected side (LW)
Ramp ascent to level walking (RA-LW)	Sound leg (RA) – affected side (RA) – sound leg (RA-LW) – affected side (RA-LW) – sound side (LW)
Ramp descent to level walking (RD-LW)	Sound leg (RD) – affected side (RD) – sound leg (RD-LW) – affected side (RD-LW) – sound side (LW)

signals without any prior knowledge. Before illustrating the proposed CNN architecture, we first give explanations of its component layers.

1) **Input Layer:** One pattern with respect to locomotion state, i.e., IMU data, is shown in Eq. (1).

$$s = [a(t_1), a(t_2), \dots, a(t_i), \dots, a(t_n)]^T \quad (1)$$

Here, $a(t_i) = (a_1(t_i), a_2(t_i), \dots, a_{M \times N}(t_i))$ is the observation vector at time t_i . M is the number of measuring units of a single IMU, and N is the total number of IMUs.

As is known, the image that is fed into the input layer for image classification typically possesses pixel matrices from multiple channels [19]. For example, the colored images in the CIFAR-10 dataset have 32×32 pixels and RGB channels. We borrow this idea and resize s as input. The sensor signal from the j th channel is firstly expressed by a $1 \times n$ vector as below:

$$x_j = (a_j(t_1), a_j(t_2), \dots, a_j(t_n))^T \quad (2)$$

where $j = 1, 2, \dots, M \times N$. Thus, each pattern X is considered as a sample with $1 \times n$ pixels and $M \times N$ channels.

2) **Convolution Layers:** Convolution layers are mainly used for feature extraction. The proposed CNN involves convolution operators with different sizes of kernels, namely, convolution kernels, which are responsible for learning relevant features from input data. The calculation of the convolution layer l is shown in Eq. (3).

$$x_j^l = f\left(\sum_{i=1}^C x_i^{l-1} * k_{ij}^l + b_j^l\right) \quad (3)$$

Here, x_i^{l-1} ($i = 1, 2, \dots, C$) is the feature map of the i th channel when pattern X goes through the layer $l - 1$. x_j^l is the output of the j th channel within the convolution layer l . The symbol $*$ stands for the convolution operator, and $f(\cdot)$ is the activation function. k_{ij}^l and b_j^l are the learnable convolution kernel matrix and bias, respectively. The output of the convolution layer l is considered as the input of the next layer in the flow.

3) **Pooling:** The purpose of pooling is to reduce the input dimensions by performing down-sampling (which also prevents overfitting of the model). More formally,

$$x_j^l = f(\beta_j^l \text{down}(x_j^{l-1}) + b_j^l), \quad (4)$$

where $\text{down}(\cdot)$ represents a sub-sampling function (max pooling, average pooling, to name a few) of the j th channel within the pooling layer l . Typically this function will operate over each distinct n -by- n block in the input map so that the output map is n -times smaller along both spatial dimensions. However, in this paper, since the input map is a one-dimensional feature map in the corresponding channel, its output is also a one-dimensional vector but has a smaller length.

4) **Fully Connected Layers:** Ordinary, fully connected layers are composed of numerous neurons. These neurons compute the weight sum of the inputs with assigned weights, and output the activations as below:

$$x_j^l = f(\omega^l x^{l-1} + b^l), \quad (5)$$

where ω^l is the weight coefficient of fully connected layers l .

Once the topology is determined, the network parameters must be learned based on the backpropagation principle. Because the error over the whole training dataset is a sum over the individual errors on each pattern, we consider minimizing the objective function as follows:

$$\text{Minimize } J(k, b, w) = \frac{1}{2} \sum_{n=1}^N \|t_n - y_n\|^2, \quad (6)$$

where t_n indicates the actual output in the form of a 1×13 vector containing the probability of each class, and y_n is the ideal output of the n th sample. For minimizing the above function, stochastic gradient descent (SGD), which slightly updates weights in the direction of the gradient, is adopted as the learning rule of the CNN.

Moreover, discovering an optimal CNN architecture usually involves a large number of parameters that significantly influence the network's performance. The most relevant parameters to determine the architecture of the CNN are the number of layers, the number and size of kernels, and the activation function in each layer. There are also some hyperparameters, such as learning rate and the mini-batch size; these are not directly related to the topology, but rather to the learning process. To summarize, the proposed CNN architecture is depicted in Fig. 3, and the parameters are listed in Table II.

As seen from Fig. 3, the architecture mainly consists of two repeated basic blocks. Each block has a convolution layer with a filter size of 1×3 at first. The stride is set as 2 so

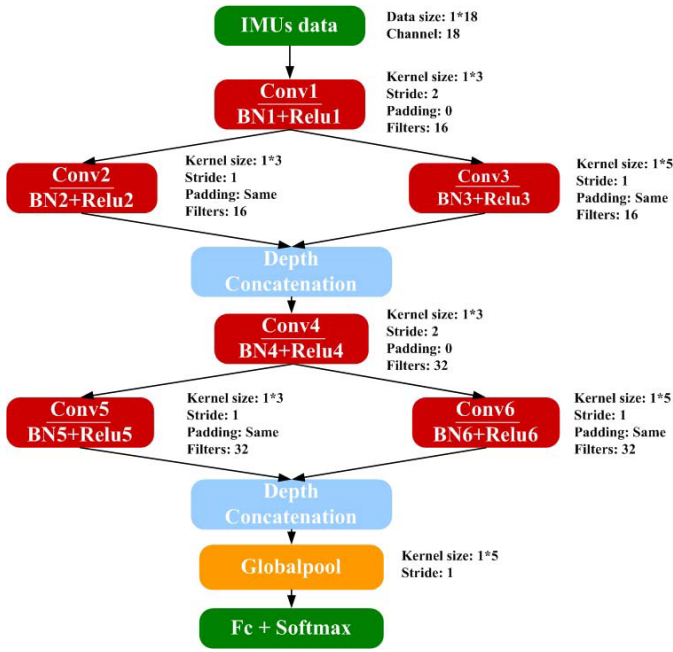


Fig. 3. Structure of the proposed CNN model.

TABLE II
HYPERPARAMETERS OF THE PROPOSED CNN MODEL

Parameter	Value
Optimizer	SGD
Mini-batch size	200
Number of epochs	50
Learning rate	0.01
Learning rate decay	0.1
Weight decay	0.0005
Momentum	0.9

that the spatial resolution of the signal is halved through the convolution. Inspired by the Inception block proposed in [36], we employ a similar structure with 1×3 and 1×5 convolutions in each block since the signals can be processed at various scales. Then, features concatenated along the channels are the input for the next layer. Additionally, to make the width and height of the information flow remain unchangeable, the stride is fixed as 1 with padding for every node within the block. Finally, we adopt global pooling [37] over feature maps in the classification layer, which is less prone to overfitting than traditional fully connected layers. For the sake of simplicity, Table III shows the details of the data flow.

Overall, the framework of our proposed method for motion intent recognition is presented in Algorithm 1.

III. EXPERIMENTS

A. Experimental Protocol

The experimental configurations are shown in Fig. 4. An 8-stair staircase with a height of 16cm was used for stair ascent/descent tests. Meanwhile, a 10° ramp approximately 6m long was used for ramp ascent/descent tests [see Fig. 4(a)]. For the devices, multiple IMUs from Noitom Perception

TABLE III
DETAILS OF THE DATA FLOW WITHIN THE MODIFIED CNN

Layer	Depth	Input size	Output size
Input	1	$1 \times 18 \times 18$	$1 \times 18 \times 18$
Conv1	1	$1 \times 18 \times 18$	$1 \times 9 \times 16$
Conv2 + Conv3	1	$1 \times 9 \times 16$	$1 \times 9 \times 32$
Conv4	1	$1 \times 9 \times 32$	$1 \times 5 \times 32$
Conv5 + Conv6	1	$1 \times 5 \times 32$	$1 \times 5 \times 64$
Global pool	1	$1 \times 5 \times 64$	$1 \times 1 \times 64$
Fc + Softmax	1	$1 \times 1 \times 64$	$1 \times 1 \times 13$
Output	1	$1 \times 1 \times 13$	1

Algorithm 1 The CNN-Based Motion Intent Recognition

Input:

The set of hyperparameters, Θ ; The set of training set, $\{X_{train}, Y_{train}\}$; The set of testing set, $\{X_{test}, Y_{test}\}$;

Output:

The recognition accuracy, acc ;

- 1: Initialize the parameters set $\Theta' = \{k, b, \omega\}$ of CNN, whose architecture is predetermined;
- 2: **while** the maximum epoch is not reached **do**
- 3: Calculate the output of the CNN through forward computation associated with input $\{X_{train}, Y_{train}\}$;
- 4: Update the Θ' by optimizing the objective function $J(k, b, \omega)$ using SGD and Θ ;
- 5: **end while**
- 6: Calculate the output of the CNN through forward computation associated with input X_{test} ;
- 7: Compute the recognition accuracy acc based on the test label Y_{test} and the network prediction T_{test} ;
- 8: **return** acc ;

Legacy¹ were employed. Each IMU consists of single triaxial accelerometer and single triaxial gyroscope with a sampling frequency of 96Hz. We only used three IMUs, and placed them at the thigh, shank, and ankle of the healthy leg [see 4(b)]. Additionally, all of the subjects were supervised by a doctor during the experiments.

Ten able-bodied subjects (five males and five females) and one transtibial amputee participated. The recruited able-bodied subjects, varied in age (18—30), height (1.58—1.83 m) and weight (40—86 kg). Similar to what Liu *et al.* did in [13], participants were required to wear a special hands-free crutch (iWalk 2.0²; see Fig. 6) for imitating the walking posture of a transfemoral amputee. The transfemoral subject, aged 67 (an amputee for 12 years), wore his own prosthesis (Teh Lin, made in Taiwan; see Fig. 5), and followed the experimental instructions (see Fig. 7).

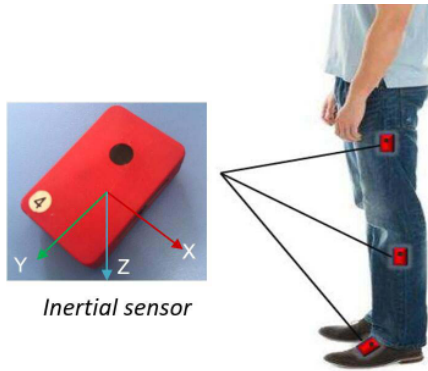
The able-bodied subjects were instructed to walk with a comfortable speed. They completed 10 trials (every trial contained at least 5 steps) of 5 steady states, as well as

¹Refer to <http://www.noitom.com/>

²Refer to <https://www.peglegs.co.uk/>



(a)



(b)

Fig. 4. Experimental configurations. (a) Experimental platform. (b) IMUs and their placement.



Fig. 5. Transtibial amputee and his prosthesis.

8 transitional states among the steady states on 5 different terrains.

The amputee subject also completed 10 repetitions of each locomotion mode (level ground, stairs, and ramp) and the necessary transitions between these locomotion modes. For the stair circuit, the subject first walked on level ground for a few strides, and then walked up stairs and transitioned back to level walking for about two strides. Then the subject stopped and turned around, walked back to the stairs, and descended. Finally, the subject continued to walk on level ground. Likewise, for the ramp circuit, the subject first walked on level ground, then walked up the ramp, and transitioned back to level walking for about two strides. Then the subject stopped and turned around, walked back to the ramp, and went down it. Finally, the subject continued walking on level ground for a few strides before stopping. We obtained 1300 samples

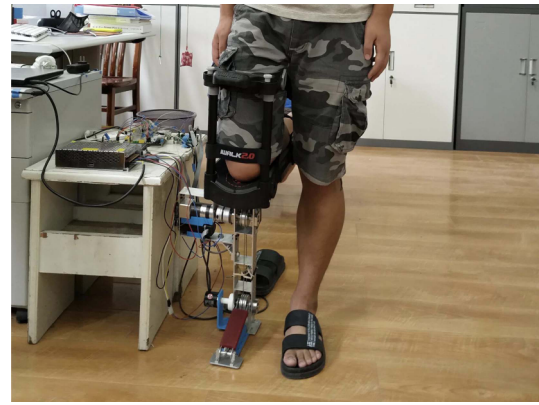


Fig. 6. An able-bodied subject wearing a hands-free crutch.



Fig. 7. Experiments of stair ascent and stair descent.

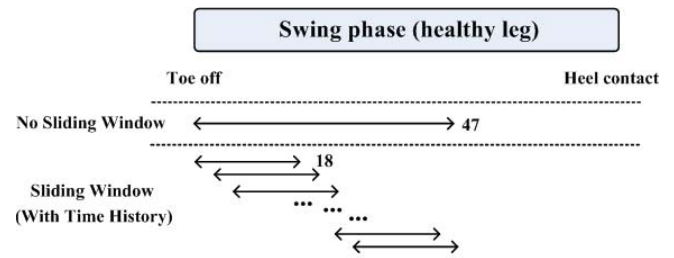


Fig. 8. Sliding window method for training data.

from the able-bodied subjects and 130 samples from the transfemoral amputee.

We did not require the amputee to follow a step sequence, and thereby the transfemoral subject had a variable number of steps during the trials. Note that transtibial amputees tend to adjust their step length before performing transition for safety. They always have the transitions from level walking to going up stairs to occur at the sound side, while transitions from level walking to going down stairs occur at the prosthesis side.

B. Data Preprocessing

The original signals were processed by a Butterworth filter with a cut-off frequency of 40 Hz. Then, IMU signals were normalized to account for the side of amputation: able-bodied subjects imitated left-sided amputees, and the transtibial amputee was a right-sided one. The side-to-side (left/right) accelerometer and the front and horizontal-plane gyroscope

TABLE IV
USER-INDEPENDENT AND USER-DEPENDENT CLASSIFICATION STRATEGIES

Classification type	Training data	Testing data	Cross validation
User independent	Nine able-bodied subjects	One novel able-bodied subject	Across subjects ($\times 10$)
User dependent	90 % pooled trials from ten able-bodied subjects	10 % withheld trial from ten able-bodied subjects	Across subjects ($\times 10$)
User dependent	80 % trials from the transtibial amputee	20 % withheld trial from the transtibial amputee	Within subject ($\times 5$)

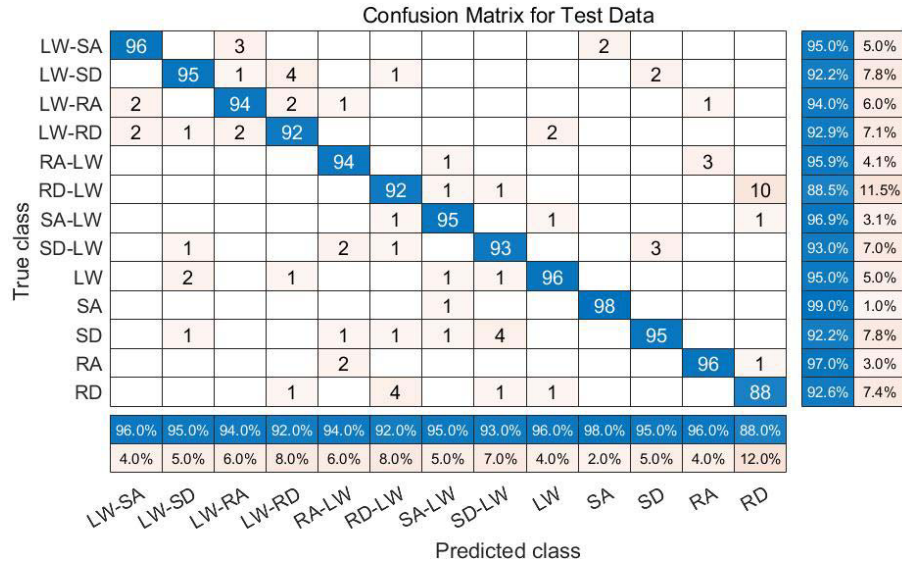


Fig. 9. Confusion matrix of user-dependent classification for able-bodied subjects.

signals were transformed (reversed sign) for subjects with left-sided amputations.

Additionally, the signals corresponding to the steady step and transitional step were extracted with a fixed window. The starting point of the window was decided by the time when the steady/transitional step began. The window size was set as 47 (about 0.5 s) while considering the sampling frequency (96 Hz). This imposes a strict restriction on the window employment, requiring the range of the extracted data to stay within the duration of the steady/transitional step. In other words, the healthy leg would be in the swing phase while the prosthesis is in the stance phase during the window. Note that the window with a size of 47 only occupies half of a whole swing phase.

We normalized the window data into $[0, 1]$ and resized the data as mentioned in Section II. Since we used three IMUs in the implementation ($M = 3$ in Eq. 1) and each IMU consists of a triaxial accelerometer and triaxial gyroscope ($N = 6$), the number of channels was fixed as 18 (computed by $M \times N$). We rescaled every data point to a 1×18 vector with 18 channels, as stated in Table III.

The sliding window method was carried out for data augmentation. The size of the sliding window was set to 18, and the stride was limited to 1 (as shown in Table 8). Thus, we imposed time history information to each processed data,

which can help improve classification performance [16]. Consequently, each sample was divided into 30 segments. Note that the data augmentation was only for training data. For matching the size of training data, we randomly extracted a window with a size of 18 from the original test sample as the final test data. Details about how to divide the dataset into training and testing data will be introduced in the following content.

C. Classification Type

User-independent and user-dependent classification cases were tested and compared in Table IV. For user-independent classification, the proposed CNN was trained on the data from nine able-bodied subjects, and tested on the data from the tenth able-bodied subject. This was repeated 10 times so that each subject was withheld once (*i.e.*, leave-one-out cross validation). To test the robustness of the algorithm, user-dependent classification was tested on both the able-bodied and amputee subjects. For able-bodied subjects, the processed sample set was randomly divided into 10 parts, where 9 parts were selected as the training set and other part was selected as the testing set. For the transtibial amputee, 20 % of samples were randomly selected as testing data, and the rest were used as training data. Cross validation was done across all 10 (5) trials such that each trial was left out once.

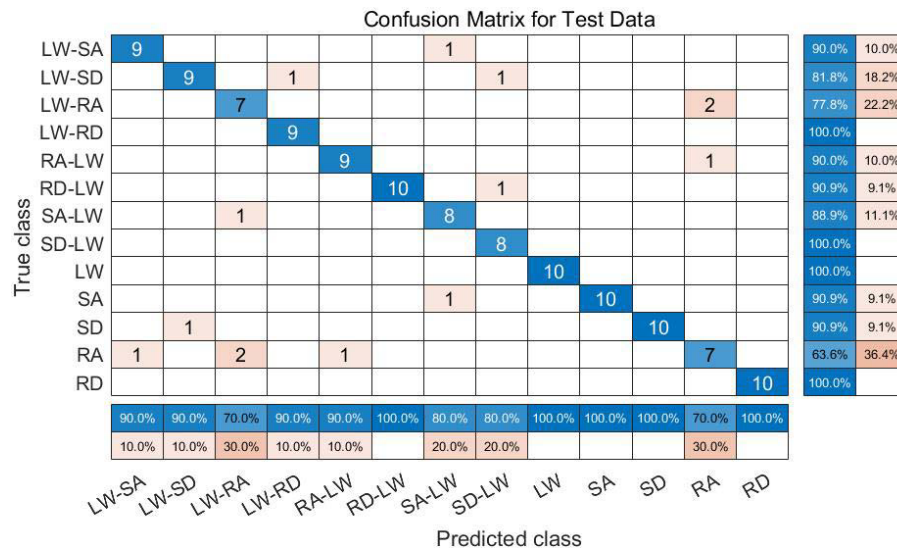


Fig. 10. Confusion matrix of user-dependent classification for transtibial amputee.

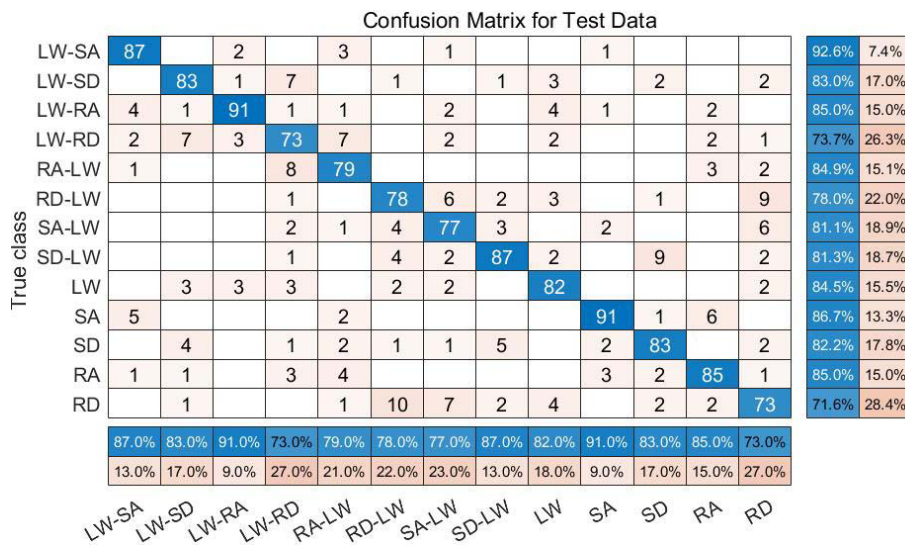


Fig. 11. Confusion matrix of user-independent classification.

D. Experimental Results

1) *User-Dependent Classification*: Fig. 9 shows the performance on user-dependent classification with respect to able-bodied subjects. The testing accuracy reached $94.15\% \pm 3.04\%$. The confusion matrix in Fig. 9 describes how many samples are misclassified within the testing set. Additionally, the numbers on the diagonal represent the correctly classified testing samples. Correctly classifying the transitional steps is much more difficult than classifying the steady steps, as reported in our previous work [23]. However, our CNN-based method addresses this disadvantage, and it can recognize both transitional steps and steady steps with high performance.

Since there is only one transtibial amputee involved, the user-dependent classification was conducted on limited samples. The dataset collected from the transtibial amputee was randomly split into training and testing data with a ratio of 8 : 2. The sliding window method was carried out on the training data; there were 26 testing data and 3120 training data in total. [Figure 10](#) shows the confusion matrix for this case.

The testing accuracy reached $89.23\% \pm 4.21\%$ on 13 motion states. The results from able-bodied subjects were basically consistent with that from the transtibial amputee, although a slight difference was observed.

2) *User-Independent Classification*: The confusion matrix among 13 motion states in the case of user-independent classification is shown in Fig. 11. The testing accuracy of user-independent classification could only reach $82.23\% \pm 7.46\%$. Correctly predicting the steady states is of great importance to the intent recognition system since the steady states are more likely to be performed in real life. Thus, we divided the experiment into two parts. One part aimed to evaluate the performance of the classifier trained by only steady state patterns, and the other evaluated the classification results that only involved transition state patterns. The results for steady states and transition states were $92.20\% \pm 4.76\%$ (Fig. 12) and $88.75\% \pm 6.68\%$, respectively, which are higher than the overall result mentioned before. The reason for this is that a large proportion of the errors are misclassifications of data

Confusion Matrix for Test Data									
True class	LW	93		1	2	7	90.3%	9.7%	
	SA		97		8		92.4%	7.6%	
	SD	1	2	96		3	94.1%	5.9%	
	RA	1	1	1	87	2	94.6%	5.4%	
	RD	5		2	3	88	89.8%	10.2%	
							93.0%	97.0%	96.0%
							7.0%	3.0%	4.0%
							87.0%	88.0%	88.0%
							13.0%	12.0%	12.0%
							LW	SA	SD
							RA	RD	
							Predicted class		

Fig. 12. Confusion matrix of steady states in user-independent classification.

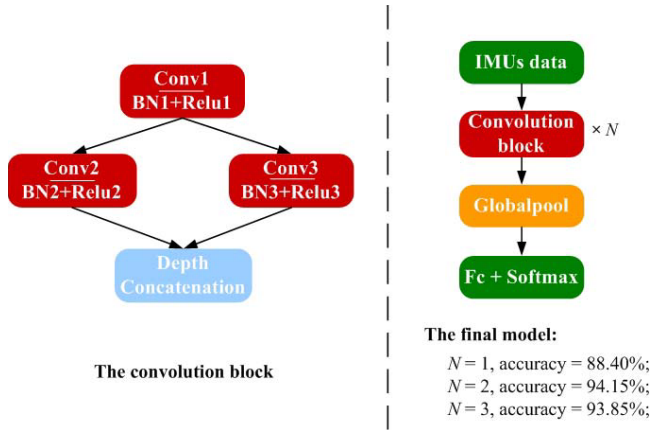


Fig. 13. Testing accuracies of the CNN with different convolution blocks.

from RD and RD-LW due to pattern similarity. Compared with Fig. 11, the classification accuracy presented in Fig. 12 increased because only 5 classes instead of all 13 classes were classified. Therefore, if independently considering steady states in the recognition system, the proposed method will make it more effective.

It seems that the misclassification rate of the user-independent system is higher than that of the user-dependent system. Thus, using subject-specific training data is highly valuable for reducing intent recognition error rates. Compared to the user-dependent strategy, the relatively high error rate on the user-independent classifier demonstrates the difficulty of generalizing training data to a fully novel user.

E. Experimental Analysis

1) *Network Architecture Search*: We incorporate a greedy-wise method for searching the optimal CNN architecture. We test 3 models, which have different numbers of convolution blocks. In Fig. 13, the left side shows the architecture of the convolution block while the right side depicts the respective performances on three cases. We can find that the model with $N = 2$ (N is the number of convolution blocks) used in this paper achieves the highest performance on user-dependent classification. This proves that adding an additional convolutional layer can lead to a better result because the testing accuracy on $N = 1$ is slightly lower than that on $N = 2$.

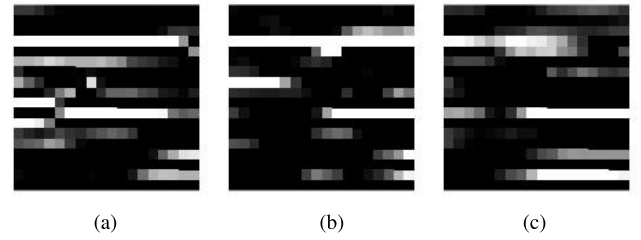


Fig. 14. Visualization of the three patterns from different classes. (a) Level walking. (b) Stair ascent. (c) Ramp ascent to level walking.

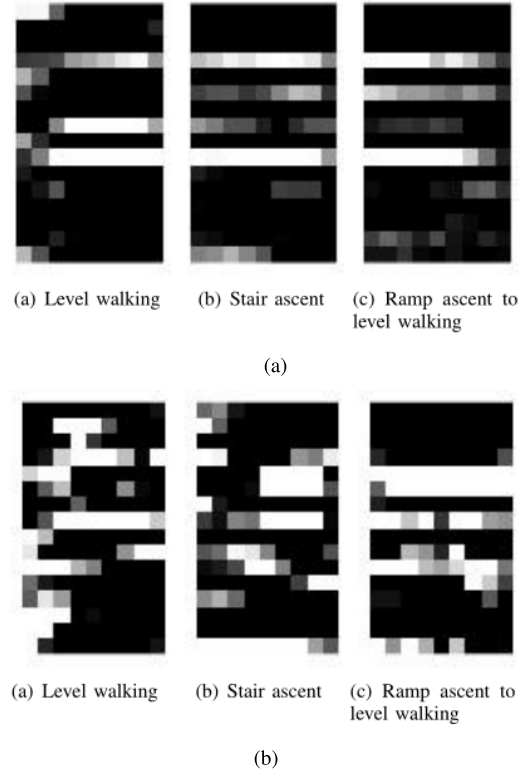


Fig. 15. Visualization of the feature maps of the first convolution layer. (a) Learning with 1 epoch. (b) Learning with 50 epochs.

Besides, to minimize computation cost, we select two convolution blocks as the final solution since the two models have similar results.

2) *Feature Visualization*: We demonstrate the validity of the feature learning procedure for intent recognition through visualization. First, we randomly select three patterns in the testing set, and visualize these samples first. Since the original 1×18 input images with 18 channels cannot be clearly described, the size of patterns is thereby set as 18×18 . The processed images are shown in Fig. 14. For any input image in the figure, the pixels in each row express the corresponding values within a specific channel. From Fig. 14, we can see that the input images are different from each other since these patterns belong to different classes. In spite of intuitive indiscernibility, these images can be learned by the deep neural network.

We further visualize the feature maps output by the first convolution layer, as shown in Fig. 15. Since the number of kernels is 16 and the size of each kernel is 1×3 in the first convolution layer, the size of output maps is designed

TABLE V
COMPARISON WITH STATE-OF-THE-ART METHODS

Reference	Subjects	Sensors	Feature extraction	Classifier	Position of sensors	Number of motion intents	Accuracy
Young et al., 2014 [16]	6 transfemoral amputees	IMU, axial load cell, motor potentiometers, encoders	Manual	DBN	Prosthesis	5	94.7 %
Young et al., 2014 [17]	6 transfemoral amputees	IMU, axial load cell, motor current	Manual	LDA	Prosthesis	13	93.9 %
Liu et al., 2017 [14]	3 able-bodied, 2 amputees	Accelerometer, gyroscope, pressure sensor	Manual	HMM	Prosthesis	5	95.8 %
Zheng et al., 2017 [6]	6 transfemoral amputees	IMU, load cell	Manual	QDA +SVM	Prosthesis	8	94.9 %
Su et al., 2018 [23]	10 able-bodied	IMUs	Manual	SVM	Healthy side	13	95.12 %
Our method	10 able-bodied 1 amputee	IMUs	Automatic	CNN	Healthy side	13	94.15 % ± 3.04 % 89.23 % ± 4.21 %

as $1 \times 9 \times 16$. Fig. 15(a) and (b) exhibit the convolution result with 1 epoch and 50 epochs training of three patterns, respectively. Although we cannot find a regular information in both results, the learning process is able to continuously update the parameters for exploring complex features. The observations show that the outputs in Fig. 15(b) have a large number of “white squares,” which demonstrates that the features become sparse with the epochs.

IV. CONCLUSIONS AND DISCUSSIONS

With the help of an intent recognition system, a powerful prosthesis allows a transfemoral amputee to perform natural transitions between neighboring locomotion modes. To improve the classical intent recognition studies, we propose a novel approach based on the CNN and IMUs data. Theoretically, it can well recognize both steady states and transitions, and provide real-time feedback to powerful prostheses. Experimental results also verify its validity and effectiveness. Table V shows the comparison between our method and state-of-the-art methods in the field of motion intent recognition. The main contributions of our proposed method can be summarized as follows.

First, we only used three IMUs rather than multi-modal sensors as conventionally used [6], [14], [16], [17]. This strategy can avoid the increased computation cost due to fusion of multi-modal signals.

Second, our approach investigated both transition states and steady states, yet some traditional works classified only steady states or transition states [6], [14], [16].

Third, we collected the IMU data from the healthy leg, which is quite different from the methodology used in past studies [6], [14], [16], [17]. Furthermore, IMU data within the early swing phase of the healthy leg was extracted and analyzed before the transition of the prosthesis occurred. This strategy was selected to explore the mapping between the motion state of the healthy leg and the transfemoral amputee’s

motion intent. Through the mapping found by machine learning, the upcoming motion state of the prosthesis can be predicted before the transition that is driven by motion intent. The difficulty of our method lies in the fact that the time series data provide little feature information for classification due to the short duration of the swing phase. As is known, the stance phase accounts for approximately 60% and the swing phase accounts for 40% within a gait cycle. Most previous studies thereby focused on the data obtained within the stance phase of the prosthesis for fully utilizing richer information. Our proposed method proves that analyzing the swing phase of the healthy leg can also realize intent recognition with high performance.

Finally, prior works often manually selected the most predictive features of signals. By contrast, our approach adopted a hierarchical CNN for automatic feature learning. This made feature engineering able to be carried out without expert participation. As a result, the new algorithm is highly competitive and superior to that in our previous work ([23]) in terms of accuracy. Note that 95.12% is the best accuracy in our previous work ([23]) while the best accuracy in this paper is 97.19%. To the authors’ best knowledge, deep learning has not yet been applied in the research field of intent recognition.

Although this study provided an alternative intent recognition system with respect to lower limb prosthesis, several limitations are present. First, this study was conducted in a laboratory environment and the able-bodied subjects were instructed to follow a pre-defined step sequence. However, a step sequence is unpredictable in real life. Furthermore, when confronted with a ramp or stairs, transfemoral amputees often transition onto the ramp with either leg (sound or prosthetic side), begin ascending stairs with their sound side, and begin descending stairs with their prosthetic side [17]. Second, whether the effect of walking speed variations can influence classification performance was not discussed in this paper. In fact, all of the subjects were instructed to walk with a comfortable speed. Finally, we tested generalization

of the proposed method to only one novel amputee. Due to the limited experimental condition, the user-independent classification was not conducted on their motion data. Future work will consider more characteristics of the step sequence of amputees, and be devoted to designing an improved intent recognition algorithm that can generalize to a range of ramp slopes and stair grades.

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